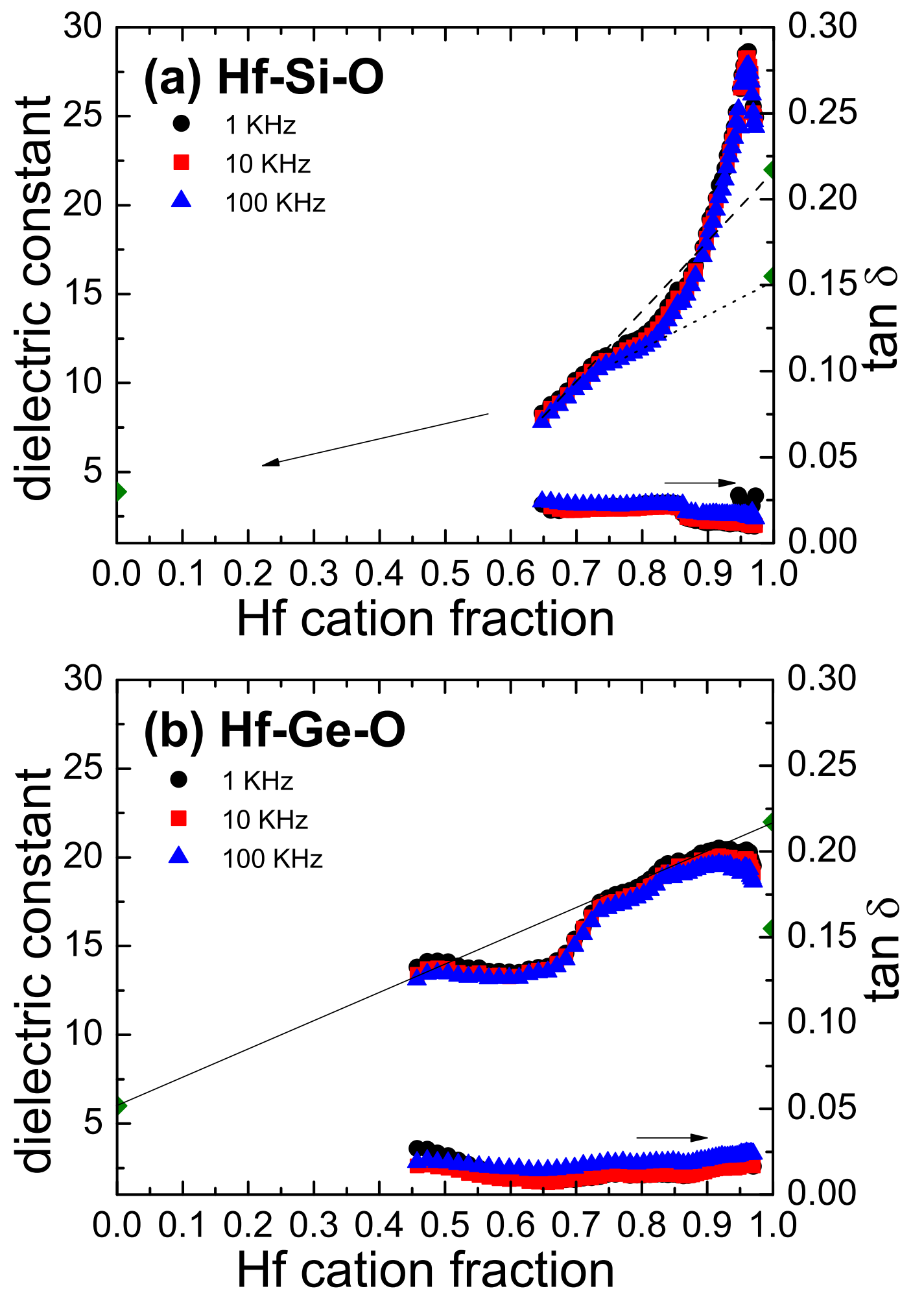

(a) Hf-Si-O
1 KHz
10 KHz
100 KHz
dielectric constant
tan δ
Hf cation fraction
(b) Hf-Ge-O
1 KHz
10 KHz
100 KHz
dielectric constant
tan δ
Hf cation fraction